

BLOCK 8

PROBABILITY AND PROBABILITY DISTRIBUTIONS

Unit 25 Probability Theory

Unit 26 Probability Distributions-I

Unit 27 Probability Distributions-II

UNIT 25 PROBABILITY THEORY

Structure

- 25.0 Objectives
- 25.1 Introduction
- 25.2 Deterministic and Non-Deterministic Experiments
- 25.3 Some Important Terminology
- 25.4 Definitions of Probability
 - 25.4.1 Classical Definition
 - 25.4.2 Axiomatic Definition
 - 25.4.3 Empirical Definition
 - 25.4.4 Subjective Definition
- 25.5 Theorems of Probability
 - 25.5.1 Theorem of Total Probability
 - 25.5.5.1 Deductions from Theorem of Total Probability
 - 25.5.2 Theorem of Compound Probability
 - 25.5.2.1 Deductions from Theorem of Compound Probability
- 25.6 Conditional Probability and Concept of Independence
 - 25.6.1 Conditional Probability
 - 25.6.2 Concept of Independent Events
- 25.7 Bayes' Theorem and its Application
- 25.8 Mathematical Expectations
- 25.9 Let Us Sum Up
- 25.10 Key Words
- 25.11 Answers/Hints to Check Your Progress Exercises
- 25.12 Exercises

25.0 OBJECTIVES

After going through this unit, you should be able to

- identify the reasons for taking decisions under uncertain situations;
- define the concept of probability;
- differentiate between total probability and compound probability;
- describe the concept of independent events; and
- solve different probability problems in accordance with theoretical prescriptions.

25.1 INTRODUCTION

Probability theory is a branch of mathematics that is concerned with random (or chance) phenomenon. It originated in the games related to chance and an Italian mathematician Jerome Cardan was the first to write on the subject. However, the basic mathematical and formal foundation on the subject was provided by Pascal and Fermat. Contribution of Russian as well as European mathematicians helped the subject to grow.

In this unit, we will learn about some basic terminologies which are used in probability related matters, deterministic and non-deterministic experiments. Further we dwell into the theorems of probability (total and compound), and conditional probability. Before moving on to the Bayes' theorem, we will understand about the concept of independence and independent events.

25.2 DETERMINISTIC AND NON-DETERMINISTIC EXPERIMENTS

The agreement among scientists regarding the validity of most scientific theories rests, to a considerable extent, on the fact that the experiment on which the theory is based will yield the same result when they are repeated. If the same results are obtained when the experiment is repeated under the same conditions, we can conclude that the results are determined by the conditions, or the experiment is deterministic. For example, anywhere in the world if we throw a stone to the sky it will certainly come back to the earth after sometime.

However, there are experiments, which do not yield the same result even if the experimental conditions are kept constant. Examples are throwing a dice or picking a card from a well-shuffled pack of cards or tossing a coin. These experiments could be thought of as the ones with unpredictable results. If we are willing to stretch our idea of experiment, then there are many examples of this kind in our day-to-day life. For example, two people living in same conditions will die at different and unpredictable ages. In literature, these experiments or events are referred to as "random experiments" or "random events". Thus, we define "non-deterministic" or "random experiments" as those *experiments* (experiment is an act which can be repeated under some given conditions) whose outcomes are not predictable before-hand. Probability and branches of statistics are developed specially to deal with this kind of random events.

Each of the following may be called a random experiment:

- 1) Tossing a coin (or several coins)
- 2) Throwing a die (or several dice)
- 3) Drawing cards from a pack
- 4) Studying the distribution of boys and girls in families of India having two children

- 5) Drawing balls from an urn (or urns) having a given number of – different-kind of balls in each.

The result of a non-deterministic event is not predictable beforehand, for there may be a number of outcomes associated with it. The same outcome of a random experiment could be described in several ways. In our 2nd example of random experiment, i.e., throwing a die, the possible outcomes are (1,2,3,4,5,6). These outcomes could have been described as “odd number of points” or “even number of points”.

The term event in probability theory is used to denote any phenomenon, which occurs as a result of a random experiment. Events can be ‘elementary’ as well as ‘composite’. An ‘elementary’ event cannot be decomposed into simpler events whereas a ‘composite’ event is an aggregate of several elementary events.

25.3 SOME IMPORTANT TERMINOLOGY

Sample Space: All possible outcomes of an on-deterministic experiment constitute the sample space of that experiment and are generally denoted by ‘S’.

For example, if one coin is tossed, either head(H) or tail (T) will appear. Therefore, H and T constitute the sample space and we can write $S = \{H, T\}$.

Similarly if two coins are tossed, $S = \{(HH), (HT), (TH), (TT)\}$

A subset of sample space, which might be favourable to the occurrence of a particular event, is called a sub-sample space. For example, in the previous example, $S_1 = \{(HH), (HT), (TH)\}$ is a sub sample space and it is favourable to the event that at least one head appears if two coins are tossed.

Example: The sample space of rolling two dice (x, y) of an experiment can be written as follows:

The number of events $N(S) = 6 \times 6 = 36$:

$$S = \{x, y\} : x = 1, 2, 3, \dots, 6; y = 1, 2, 3, \dots, 6$$

Here x and y represent the number turned up in the top of the first and second dice respectively.

Events: When some of the elements of all the possible outcomes in the sample space of a random experiment satisfy a particular criterion, we call it an event. If three unbiased coins are tossed, the sample space is

$$S = \{(HHH), (HHT), (HTH), (HTT), (THH), (THT), (TTH), (TTT)\} \text{ and}$$

Following events can be picked up from it:

$$E_1 = \{(HHH)\} = \text{Event of getting all heads}$$

$$E_2 = \{(HHH), (HHT), (HTH), (THH)\} = \text{Event of getting at least 2 heads}$$

$$E_3 = \{(HHT), (HTH), (THH)\} = \text{Event of getting exactly 2 heads} \dots \dots \dots \text{etc.}$$

Example: Consider an event A that the total number of points rolled with a pair of dice is 7. So among the 36 possibilities only (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) yield a total of 7. So we write:

$$A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$$

Thus an event (outcome or result) can be identified as a collection of points which constitute a subset of appropriate sample space for which the particular event occurs.

Mutually Exclusive Events: Events are said to be mutually exclusive if two or more of them cannot occur simultaneously. For example, while tossing a coin, the elementary events 'head' and 'tail' are mutually exclusive. Similarly, when we throw a die the appearance of the numbers 1, 2, 3, 4, 5, 6 are mutually exclusive. We define the events from these experiments as follows:

A: the event of odd number of points; and

B: the event of even number of points.

These are also mutually exclusive

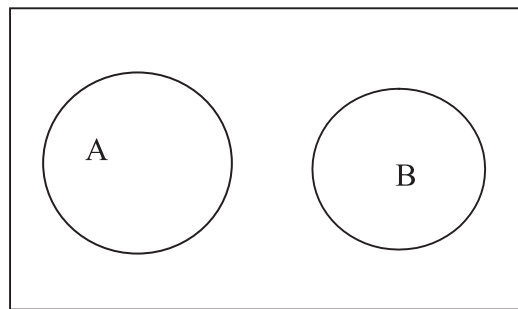


Fig. 25.1: Mutually Exclusive Events

In Fig. 25.1, the rectangular box represents the sample space and the circles represented by A and B represent subsample spaces and contain favourable elements to the events A and B. The complete separation of the circles indicates that there is no element, which is common to both the events. Thus, A and B events are mutually exclusive. The above way of representing events is called Venn diagram.

Mutually Exhaustive Events: Several events are said to be mutually exhaustive if and only if at least one of them necessarily occurs. For example, while tossing a coin, the events of head and tail are mutually exhaustive, as one of them must occur.

Equally Likely Events: The outcomes of a non-deterministic event are said to be equally Likely if occurrence of none of them can be expected in preference to another. For example, while tossing a coin, the occurrence of the event head or tail is equally likely if the coin is unbiased.

Exhaustive Events: Several events are said to form an exhaustive set, if atleast one of them must necessarily occur. The complete group of all elementary events of a random experiment given is an exhaustive set of events. For example, in tossing a coin the two events head or tail form an exhaustive set because one of these two must necessarily occur.

Independent Events: Events are said to be independent of each other if occurrence of one event is not affected by the occurrence of the others. For example, while throwing a die repeatedly, the event of getting a '3' in the first throw is independent of getting a '6' in the second throw.

Conditional Events: When events are neither independent nor mutually exclusive, it is possible to think that one of them is dependent on the other. For example, it may or may not rain if the day is cloudy but if there is rain, there must be clouds in the sky. Thus, the event of rain is conditioned upon the event of clouds in the sky.

25.4 DEFINITIONS OF PROBABILITY

The term probability has been interpreted in terms of four definitions viz.,

- 1) Classical definition
- 2) Axiomatic definition
- 3) Empirical definition
- 4) Subjective definition

25.4.1 Classical Definition

The classical definition states that if an experiment consists of N outcomes which are mutually exclusive, exhaustive and equally likely and N_A are the number of outcomes that are favourable to an event A , then the probability of the event A ($P(A)$) is defined as

$$P(A) = N_A / N$$

In other words, the probability of an event A equals the ratio of the number of outcomes N_A favourable to A to the total number of outcomes. See the following example for a better understanding of the concept.

Example: Two unbiased dice are thrown simultaneously. Find the probability that the product of the points appearing on the dice is 18.

There are 36 (N) possible outcomes if two dice are thrown simultaneously. These outcomes are mutually exclusive, exhaustive and equally likely based on the assumption that the dice are unbiased. Now we denote A : the product of the points appearing on the dice is 18.

The events favourable to ' A ' are $[(3, 6), (6, 3)]$ only, therefore, $N_A = 2$. According to classical definition of probability

$$P(A) = N_A / N = 2 / 36 = 1 / 18$$

When none of the outcome is favourable to the event A , $N_A = 0$, $P(A)$ also takes the value 0. In that case we say that event A is impossible. If $N_A = N$, $P(A)$ takes the value 1. In that case we say that event A is a sure event.

There are some limitations of the classical definition of probability. Unless the outcomes of an event are mutually exclusive, exhaustive and equally likely, classical definition cannot be applied. Again, if the number of outcomes of an event is infinitely large, the definition fails. The phrase ‘equally likely’ appearing in the classical definition of probability means equally probable, thus the definition is circular in nature.

Techniques of Counting

- 1) Fundamental Principle of Counting if several processes can be performed in the following manner: the first process in p ways, the second in q ways, the third in r ways, and so on, then the total number of ways in which the whole process can be performed in the order indicated is given by their product: $p \cdot q \cdot r \dots$

- 2) Permutation: The total number of ways of arranging n distinct objects taken r at a time is given by: ${}^n P_r$

$${}^n P_r = n(n-1)(n-2)\dots(n-r+1)$$

- 3) Arrangement in a line or circle: The total number of ways in which n distinct objects can be arranged among themselves is:

- i) in a line: $n! = 1 \cdot 2 \cdot 3 \cdot 4 \dots n$

- ii) in a circle: $(n-1)!$

- 4) Permutation with repetition: The number of ways of arranging n objects among which p are alike, q are alike, r are alike, etc. is: $\frac{n!}{p! \cdot q! \cdot r! \dots}$

- 5) Combination: The total number of possible groups that can be formed by taking r objects out of n distinct objects is given by: ${}^n C_r =$

$${}^n C_r = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}$$

- 6) Choosing balls from an urn: The total number of ways of choosing a number of white balls and b number of black balls from an urn containing A number of white balls and B number of black balls is: ${}^A C_a \cdot {}^B C_b$

- 7) Ordered Partition (Distinct Objects): The total number of ways of distributing n distinct objects into r components marked as 1, 2, 3, 4, ..., r is r^n

The number of ways in which n objects can be distributed so that the r compartments contain respectively $n_1, n_2, n_3, \dots, n_r$ objects is

$$\frac{n!}{n_1! n_2! n_3! \dots n_r!}$$

- 8) Ordered Partition (identical objects): The total number of ways of distributing n identical objects into r compartments marked as 1, 2, 3, ..., r is ${}^{n+r-1}C_{r-1}$

If none of the compartments should remain empty, the total number of ways of distributing the objects is ${}^{n-1}C_{r-1}$

25.4.2 Axiomatic Definition

In the axiomatic definition of probability, we start with a probability space 'S' where set 'S' of abstract objects is called outcomes. The set S and its subsets are called events. The probability of an outcome A is by definition a number $P(A)$ assigned to A. Such a number satisfies the following axioms:

- $P(A) \geq 0$ i.e., $P(A)$ is a non-negative number.
- The probability of the certain event S is 1, i.e., $P(S)=1$.
- If two events A and B have no common elements, or, A and B are mutually exclusive, the probability of the event consisting of the outcomes that are in A or in B equals to sum of their probabilities, which can be written as A union B (AUB):

$$P(A \cup B) = P(A) + P(B)$$

We comment next on the connection between an abstract sample space and the underlying real experiment. The first step in model formation is between elements of S and its experimental outcomes. The actual outcomes of a real experiment can involve a large number of observable characteristics. In the formation of the model, we select from these characteristics the one that is of interest in our investigation.

For example, consider the possible models of the throwing of an unbiased die by the 3 players X, Y and Z.

X says that the outcomes of this consist of six faces of the die, forming the sample space $\{1, 2, 3, 4, 5, 6\}$.

Y argues that the experiment has only 2 outcomes, even or odd, forming the sample space $\{\text{even}, \text{odd}\}$

Z bets that the die will rest on the left side of the table and the face with one point will show. Her experiment consists of infinitely many points consisting of the six faces of the die and the coordinate of the table where the die rests finally.

25.4.3 Empirical Definition

In N trials of a random experiment, if an event is found to occur m times, the relative frequency of the occurrence of the event is m/N . If this relative frequency approaches a limiting value p , as N increases indefinitely, then 'p' is called the probability of the event A.

$$P(A) = \lim_{N \rightarrow \infty} \left(\frac{m}{N} \right)$$

To give a meaning to the limit we must interpret the above formula as an assumption used to define $P(A)$. This concept was introduced by Von Mises. However, the use of such a definition as a basis of deductive theory has not enjoyed wide acceptance.

25.4.4 Subjective Definition

In subjective interpretation of probability, the number $P(A)$ is assigned to a statement. A , is a measure of our state of knowledge or belief concerning the truth of A . These kinds of probabilities are most often used in our daily life and conversations. We often make statements like “I am 100% sure that I will pass the examination” i.e., $P(\text{of passing the examinations}) = 1$, or “there is 50% chance that India will win the match against Pakistan” i.e., $P(\text{India will win the match against Pakistan}) = \frac{1}{2}$

Check Your Progress 1

- 1) What is the probability that all three children born in a family will have different birthdays? Hint: Use rule number 1 (Techniques of Counting section)

.....

- 2) Five persons a, b, c, d, e occupy seats in a row at random. What is the probability that a and b will sit next to each other? Hint: Use rule number 3 (Techniques of Counting section)

.....

- 3) Two cards are drawn at random from a pack of well-shuffled cards. Find the probability that

- a) Both cards are red
- b) One is a heart and another is a diamond

.....

